

# RETAIL SALES ANALYSIS DASHBOARD USING EXCEL AND POWER BI

## 1. Project Overview and Objective:

### Overview:

This project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

### Objective:

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

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## 2. Data Sources:

- **Source Description and Timeline:** Google Dataset Search **and** 2020-2024.
  - **Domain:** Retail Sales Analytics
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## 3. Problem Statement:

- To analyse sales performance across regions and products to identify top-performing areas and profit trends.
  - To understand customer purchasing behavior to enhance marketing strategies.
  - **Goal:** Build an automated Power BI dashboard backed by clean Excel data to track sales trends, monitor stock levels, and cut reporting time by 80%.
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## 4. Attribute (Column /Features) Details:

| Attribute Name  | Data Type       | Description                         |
|-----------------|-----------------|-------------------------------------|
| Customer_ID     | Integer/String  | Unique identifier for each Customer |
| Gender          | Text            | It represents Male or Female        |
| Region          | String(Text)    | Region of the Customer              |
| Age             | Numeric         | Age of the Customer in years        |
| Product Name    | String(Text)    | Individual Name of the Product      |
| Category        | String(Text)    | Categories of the Product           |
| Unit Price      | Numeric         | Price of the single Piece           |
| Quantity        | Numeric         | Number of Products Buying           |
| Total Price     | Numeric         | Amount of Products                  |
| Shipping Fee    | Numeric         | Shipping cost of the Product        |
| Shipping Status | Integer(String) | Current Stage of Shipment           |



The Missing Values of Age Column is Handled using Function =MEDIAN(D2:D1001) as follows, and value is detected and filled with the value 49.

|       |                                   |        |       |     |            |
|-------|-----------------------------------|--------|-------|-----|------------|
| C1011 |                                   |        |       |     |            |
|       | A                                 | B      | C     | D   | E          |
| 992   | CUST0298                          | Female | East  | 58  | Mouse      |
| 993   | CUST0299                          | Male   | East  | 58  | Keyboard   |
| 994   | CUST0299                          | Male   | West  | 58  | Headphones |
| 995   | CUST0299                          | Female | West  | 58  | Smartwatch |
| 996   | CUST0299                          | Female | East  | 58  | Smartphone |
| 997   | CUST0299                          | Male   | West  | 58  | Smartphone |
| 998   | CUST0299                          | Female | South | 58  | Mouse      |
| 999   | CUST0299                          | Female | North | 58  | Smartwatch |
| 1000  | CUST0300                          | Female | East  |     | Mouse      |
| 1001  | CUST0300                          | Male   | East  | 66  | Smartwatch |
| 1002  |                                   |        |       |     |            |
| 1003  |                                   |        |       |     |            |
| 1004  |                                   |        |       |     |            |
| 1005  | Total Count Missing Values        |        |       | 224 |            |
| 1006  | Count Of Missing Value in Age     |        |       | 102 |            |
| 1007  | Missing values in Category        |        |       | 2   |            |
| 1008  | Missing values in Region          |        |       | 52  |            |
| 1009  | Missing Values in Shipping Status |        |       | 52  |            |
| 1010  |                                   |        |       | 0   |            |
| 1011  | Median for Age Column             |        |       | 49  |            |

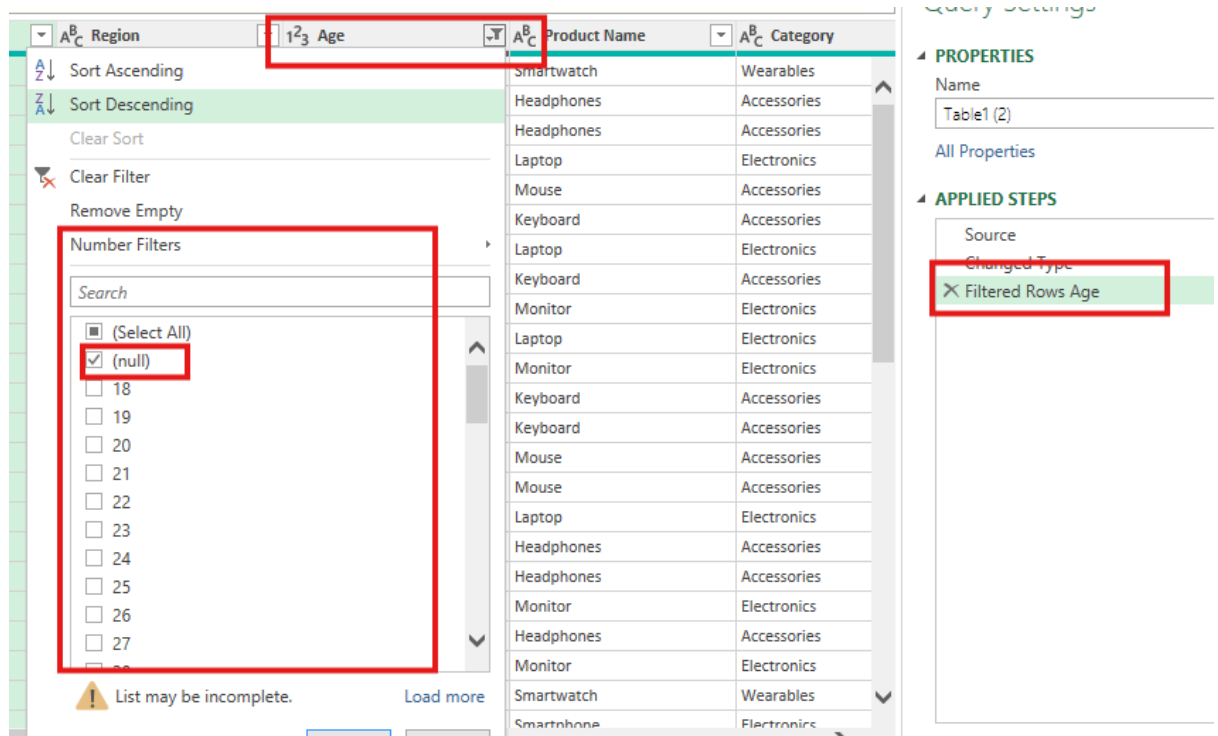
### Transformation:

The above image represents the Renames of Order Date3 as Order Year and Order Date1, Order Date2 as Renamed by Order Date, Order Month respectively. By Right Clicking in the Column Name we can choose Rename then give which name we want. The Transformation is simple and Effective for the Retail sale Data.

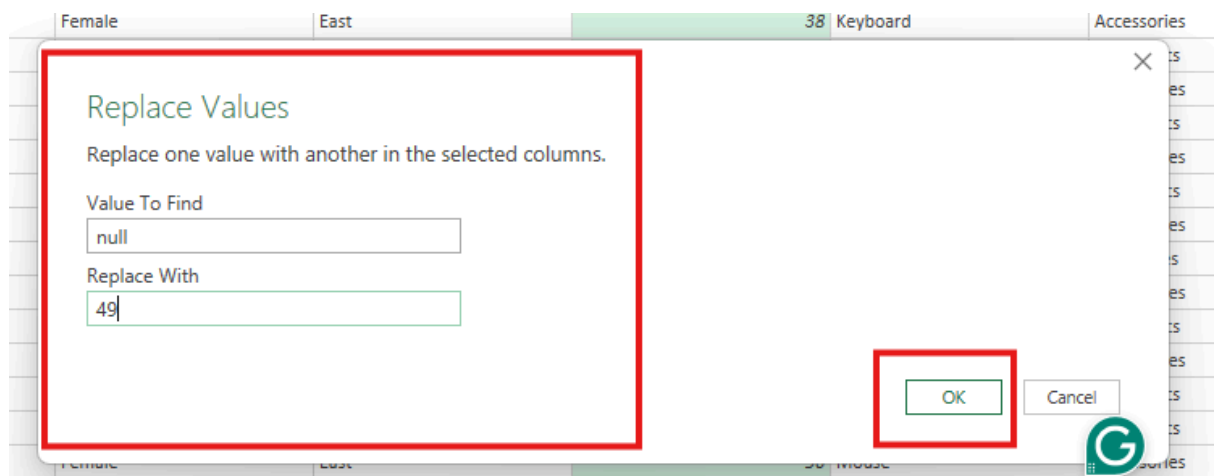
| = Table1.RenameColumns(#"Changed Type1",{"Order Date.3.2", "Order Duration"}, {"Order Date.3.1", "Order Year"}, |              |            |             |            |                |  |
|-----------------------------------------------------------------------------------------------------------------|--------------|------------|-------------|------------|----------------|--|
| Order Fee                                                                                                       | Order Status | Order Date | Order Month | Order Year | Order Duration |  |
| 12                                                                                                              | Delivered    | 18         | 1           | 2023       | 00:00          |  |
| 9                                                                                                               | Delivered    | 14         | 2           | 2023       | 00:00          |  |
| 11                                                                                                              | Delivered    | 3          | 1           | 2024       | 00:00          |  |
| 6                                                                                                               | Delivered    | 18         | 2           | 2023       | 00:00          |  |
| 14                                                                                                              | Delivered    | 2          | 8           | 2023       | 00:00          |  |
| 14                                                                                                              | Delivered    | 7          | 4           | 2023       | 00:00          |  |
| 16                                                                                                              | Delivered    | 10         | 4           | 2023       | 00:00          |  |
| 16                                                                                                              | Delivered    | 2          | 2           | 2023       | 00:00          |  |
| 14                                                                                                              | Delivered    | 29         | 5           | 2023       | 00:00          |  |
| 19                                                                                                              | Delivered    | 8          | 10          | 2023       | 00:00          |  |
| 7                                                                                                               | Delivered    | 27         | 3           | 2023       | 00:00          |  |
| 16                                                                                                              | Delivered    | 21         | 4           | 2023       | 00:00          |  |
| 14                                                                                                              | Delivered    | 6          | 6           | 2023       | 00:00          |  |
| 14                                                                                                              | Delivered    | 4          | 11          | 2023       | 00:00          |  |
| 19                                                                                                              | Delivered    | 18         | 12          | 2023       | 00:00          |  |
| 13                                                                                                              | Delivered    | 12         | 3           | 2023       | 00:00          |  |
| 14                                                                                                              | Delivered    | 11         | 2           | 2023       | 00:00          |  |
| 7                                                                                                               | Delivered    | 13         | 8           | 2023       | 00:00          |  |
| 6                                                                                                               | Delivered    | 21         | 7           | 2023       | 00:00          |  |
| 17                                                                                                              | Delivered    | 10         | 1           | 2023       | 00:00          |  |
| 19                                                                                                              | Delivered    | 3          | 10          | 2023       | 00:00          |  |
| 19                                                                                                              | Delivered    | 28         | 7           | 2023       | 00:00          |  |
| 19                                                                                                              | Delivered    | 17         | 12          | 2023       | 00:00          |  |

- **Filtering & Sorting:** Organized data to focus on relevant records.

Filtering And Sorting has in the Power Query Excel Editor as follows,

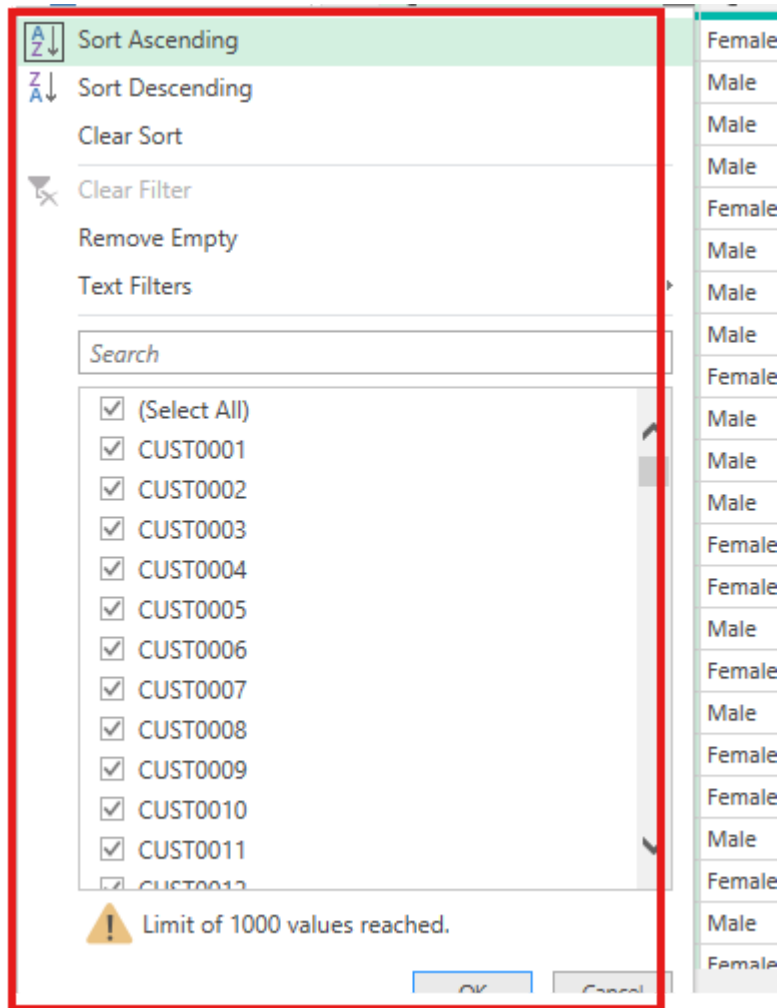


Filtering the Age Column by Right Clicking the Age Column Name by using filter i have selected the null to count how much is there in the Age Column data. By using the Replace values in Home Tab i have replaced the Values as Follows,



### Sorting:

Sorting the Certain Range in a Table to Find Values as we require to Find Exact Data. For Example Customer ID if it is in Unarranged order it will be difficult to find Exact Data to Show so we Sorting as Ascending or Descending as we Require for Prediction.



Here the sorting is done for CUSTOMER ID as CUSTID0001,CUSTID0002,.....,CUSTID1000 using DATA TAB—> In Sort & Filter—>Select Sort(Ascending) for Customer ID we want ascending Only so i have Choose it.Then the sorting is complete.

### Round-off Unit Price,Shipping fee

Table: TransformColumns(#"Rounded Off Unit Price",{"Shipping Fee",each Number.Round(\_, 0), type number})

|    | 1.2 Unit Price | 1.3 Quantity | 1.3 Total Price | 1.2 Shipping Fee | 1.3 Shipping Status | 1.3 Order Date |
|----|----------------|--------------|-----------------|------------------|---------------------|----------------|
| 1  | 50             | 3            | 150             | 6                | Returned            | 10             |
| 2  | 1500           | 4            | 6000            | 11               | Returned            | 26             |
| 3  |                |              |                 |                  |                     | 18             |
| 4  |                |              |                 |                  |                     | 10             |
| 5  |                |              |                 |                  |                     | 18             |
| 6  |                |              |                 |                  |                     | 06             |
| 7  |                |              |                 |                  |                     | 14             |
| 8  |                |              |                 |                  |                     | 07             |
| 9  |                |              |                 |                  |                     | 06             |
| 10 |                |              |                 |                  |                     | 03             |
| 11 |                |              |                 |                  |                     | 04             |
| 12 |                |              |                 |                  |                     | 14             |
| 13 |                |              |                 |                  |                     | 29             |
| 14 | 30             | 2            | 60              | 8                | In Transit          | 15             |
| 15 | 1500           | 5            | 7500            | 20               | In Transit          | 17             |
| 16 | 300            | 3            | 900             | 6                | Delivered           | 18             |
| 17 | 200            | 1            | 200             | 14               | Delivered           | 02             |
| 18 | 200            | 3            | 600             | 14               | Delivered           | 07             |
| 19 | 100            | 2            | 200             | 19               | In Transit          | 24             |
| 20 | 200            | 5            | 1000            | 13               | In Transit          | 15             |
| 21 | 100            | 2            | 200             | 12               | In Transit          | 28             |
| 22 | 100            | 3            | 300             | 20               | In Transit          | 16             |
| 23 | 200            | 3            | 600             | 16               | Delivered           | 10             |

Round dialog box: Specify how many decimal places to round to. Decimal Places: 0

APPLIED STEPS: Source, Changed Type, Rounded Off Unit Price, Rounded Off Shipping fee

Here the Shipping fee and Unit Price Column Rounded of First Select that Column and Using Home Tab→Select Transform tools—in that select Data Type —Click Decimal Number Dialog box will open in that give Decimal Points.I have given 0 as Decimal Point so i got the Whole Number as My Unit Price,Shipping Fee Columns as respectively.

Query Settings

PROPERTIES

Name

Table1

All Properties

APPLIED STEPS

Source

Changed Type

Rounded Off

Table

Category

1.2 Unit Price

Quantity

Total Price

1.2 Shipping Fee

Shipping Fee

Shipping Status

Order Date

Order Month

Order Year

Order Duration

## Renamed Column Name

Query Settings

PROPERTIES

Name

Table1

All Properties

APPLIED STEPS

Source

Changed Type

Rounded Off Unit Price

Rounded Off Shipping fee

Replaced Value

Filtered Rows

Split Column by Delimiter

Split Column Order date

Split Column by Delimiter1

Changed Type1

Renamed Columns Order Year

Shipping Fee

Shipping Status

Order Date

Order Month

Order Year

Order Duration

Created a new Column named as Year, Month, Date for Checking Month,Year Order Details Independently.The New Column are Created Using Splitting Column from Home Tab in that Select Delimiter its separates using “-” to Separate Data.

- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights. Pivot is an easy tool to summarise data in excel.

Here the Pivot table of Retail Sales Analysis

|                          |       |                 |          |          |          |          |                    |
|--------------------------|-------|-----------------|----------|----------|----------|----------|--------------------|
| Shipping Fee             | (All) |                 |          |          |          |          |                    |
| <b>Sum of Order Date</b> |       | <b>Quantity</b> |          |          |          |          |                    |
| <b>Shipping Status</b>   |       | <b>1</b>        | <b>2</b> | <b>3</b> | <b>4</b> | <b>5</b> | <b>Grand Total</b> |
| Delivered                |       | 1159            | 924      | 1119     | 1197     | 964      | 5363               |
| In Transit               |       | 903             | 805      | 1308     | 1221     | 924      | 5161               |
| Returned                 |       | 1075            | 1019     | 1011     | 784      | 898      | 4787               |
| Grand Total              |       | 3137            | 2748     | 3438     | 3202     | 2786     | 15311              |

|                           |               |               |                    |  |
|---------------------------|---------------|---------------|--------------------|--|
| <b>Category</b>           | (All)         |               |                    |  |
| <b>Sum of Total Price</b> |               | <b>Gender</b> |                    |  |
| <b>Product Name</b>       | <b>Female</b> | <b>Male</b>   | <b>Grand Total</b> |  |
| Headphones                | 17000         | 20000         | 37000              |  |
| Keyboard                  | 10950         | 11950         | 22900              |  |
| Laptop                    | 331500        | 364500        | 696000             |  |
| Monitor                   | 66300         | 84600         | 150900             |  |
| Mouse                     | 6660          | 5340          | 12000              |  |
| Smartphone                | 166400        | 187200        | 353600             |  |
| Smartwatch                | 32200         | 42000         | 74200              |  |
| Grand Total               | 631010        | 715590        | 1346600            |  |
|                           | Shipping Fee  |               | (All)              |  |

Here the Total price of the Product is Specified using Gender and the High Byer of the Product is Male. If have more Products Category of Male Product will consume more Revenue.

|                        |                     |                 |               |                |              |                   |                   |                    |  |  |
|------------------------|---------------------|-----------------|---------------|----------------|--------------|-------------------|-------------------|--------------------|--|--|
| <b>Category</b>        | (All)               |                 |               |                |              |                   |                   |                    |  |  |
| <b>Sum of Quantity</b> | <b>Product Name</b> |                 |               |                |              |                   |                   |                    |  |  |
| <b>Shipping Status</b> | <b>Headphones</b>   | <b>Keyboard</b> | <b>Laptop</b> | <b>Monitor</b> | <b>Mouse</b> | <b>Smartphone</b> | <b>Smartwatch</b> | <b>Grand Total</b> |  |  |
| Delivered              | 138                 | 178             | 176           | 175            | 140          | 148               | 128               | 1083               |  |  |
| In Transit             | 112                 | 140             | 139           | 182            | 122          | 166               | 129               | 990                |  |  |
| Returned               | 120                 | 140             | 149           | 146            | 138          | 128               | 114               | 935                |  |  |
| Grand Total            | 370                 | 458             | 464           | 503            | 400          | 442               | 371               | 3008               |  |  |

|                          |              |
|--------------------------|--------------|
| <b>Sum of Unit Price</b> |              |
| <b>Region</b>            | <b>Total</b> |
| East                     | 104607.6     |
| North                    | 109647.2     |
| South                    | 103755.7     |
| West                     | 139693.4     |
| Grand Total              | 457703.9     |

|                             |              |
|-----------------------------|--------------|
| <b>Count of Customer ID</b> |              |
| <b>Region</b>               | <b>Total</b> |
| East                        | 231          |
| North                       | 229          |
| South                       | 244          |
| West                        | 296          |
| Grand Total                 | 1000         |

Data shows the Total No of Customer Persist in the Dataset as classified as Region.

## 7. Data Modelling and DAX (Power BI)

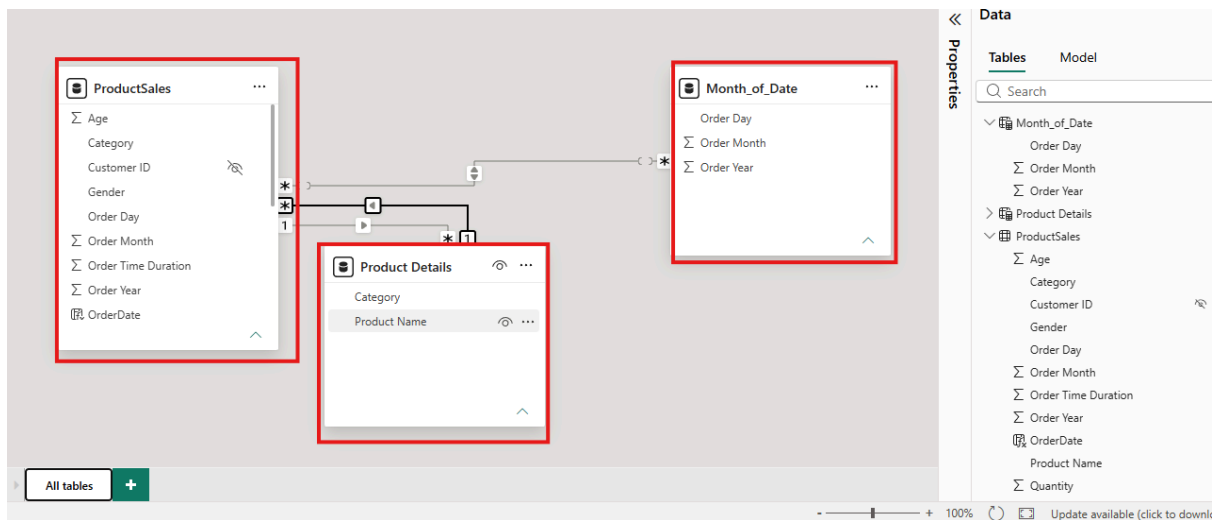
- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

Merge Query is used for establishing the relationship between tables, here my dataset of Retail Sales Analysis has only one table so, I have done this cardinality in the Power BI for that table only. I have created the Calculated Table inside in the table as Product Details, Month of date.

### Data Model for Retail sales:

Here the Relationship between Product Sales table and Product details is defined by Cardinality as **One to Many**. And the Relationship Between Month of date table and Product Sales Many to Many.





## SCREENSHOT OF DATA MODELLING

- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals. **(List Out)**

### Calculated Column

In Power BI The Excel data file loaded using BlankDocument →Get data→Select required saved excel file and check the preview it is in comma separator and Click Load in Power BI. Then the Power Query will open in that the calculations have been done.

| Total Product Price = ProductSales[Unit Price] * ProductSales[Quantity] + ProductSales[Shipping Fee] |          |             |              |                 |            |             |            |                     |                     |  |
|------------------------------------------------------------------------------------------------------|----------|-------------|--------------|-----------------|------------|-------------|------------|---------------------|---------------------|--|
| Unit Price                                                                                           | Quantity | Total Price | Shipping Fee | Shipping Status | Order Date | Order Month | Order Year | Order Time Duration | Total Product Price |  |
| 300                                                                                                  | 3        | 900         | 9            | In Transit      | 14         | 2           | 2023       | 0                   | 909                 |  |
| 300                                                                                                  | 3        | 900         | 6            | Delivered       | 18         | 2           | 2023       | 0                   | 906                 |  |
| 300                                                                                                  | 2        | 600         | 8            | In Transit      | 10         | 1           | 2023       | 0                   | 608                 |  |
| 300                                                                                                  | 1        | 300         | 10           | Returned        | 30         | 3           | 2023       | 0                   | 310                 |  |
| 300                                                                                                  | 5        | 1500        | 13           | Delivered       | 12         | 3           | 2023       | 0                   | 1513                |  |
| 300                                                                                                  | 2        | 600         | 14           | Delivered       | 11         | 2           | 2023       | 0                   | 614                 |  |
| 300                                                                                                  | 1        | 300         | 7            | Returned        | 18         | 12          | 2023       | 0                   | 307                 |  |
| 300                                                                                                  | 1        | 300         | 7            | In Transit      | 14         | 1           | 2023       | 0                   | 307                 |  |
| 300                                                                                                  | 2        | 600         | 17           | Delivered       | 16         | 4           | 2023       | 0                   | 617                 |  |
| 300                                                                                                  | 1        | 300         | 15           | Delivered       | 5          | 4           | 2023       | 0                   | 315                 |  |
| 300                                                                                                  | 1        | 300         | 7            | Delivered       | 26         | 10          | 2023       | 0                   | 307                 |  |
| 300                                                                                                  | 3        | 900         | 18           | Delivered       | 20         | 7           | 2023       | 0                   | 918                 |  |
| 300                                                                                                  | 5        | 1500        | 9            | In Transit      | 30         | 10          | 2023       | 0                   | 1509                |  |
| 300                                                                                                  | 2        | 600         | 16           | Returned        | 2          | 9           | 2023       | 0                   | 616                 |  |
| 300                                                                                                  | 3        | 900         | 17           | Returned        | 22         | 5           | 2023       | 0                   | 917                 |  |
| 300                                                                                                  | 1        | 300         | 9            | Returned        | 25         | 7           | 2023       | 0                   | 309                 |  |
| 300                                                                                                  | 1        | 300         | 17           | In Transit      | 23         | 11          | 2023       | 0                   | 317                 |  |
| 300                                                                                                  | 2        | 600         | 18           | Returned        | 19         | 3           | 2023       | 0                   | 618                 |  |
| 300                                                                                                  | 1        | 300         | 14           | Delivered       | 5          | 8           | 2023       | 0                   | 314                 |  |
| 300                                                                                                  | 4        | 1200        | 15           | In Transit      | 28         | 1           | 2023       | 0                   | 1215                |  |
| 300                                                                                                  | 1        | 300         | 17           | In Transit      | 3          | 4           | 2023       | 0                   | 317                 |  |
| 300                                                                                                  | 5        | 1500        | 18           | Returned        | 21         | 3           | 2023       | 0                   | 1518                |  |

Using Dax formula Select Table Tools→ Calculations Column —Select New Column to create new column as Total Product Price the syntax is,

**Total Product Price = ProductSales[Unit Price] \* ProductSales[Quantity] + ProductSales[Shipping Fee]**

The screenshot shows the Power BI Desktop interface. The 'Formulas' tab is active, displaying the DAX formula for the 'OrderDate' column: `OrderDate = CONCATENATE(CONCATENATE(CONCATENATE(ProductSales[Order Day],"/"),ProductSales[Order Month]),CONCATENATE("/","Year"))`. The 'Data' pane on the right shows the 'ProductSales' table with columns: Quantity, Total Price, Shipping Fee, Shipping Status, Order Day, Order Month, Order Year, Order Time Duration, Total Product Price, and OrderDate. The 'OrderDate' column is highlighted in the 'Data' pane.

| Quantity | Total Price | Shipping Fee | Shipping Status | Order Day | Order Month | Order Year | Order Time Duration | Total Product Price | OrderDate  |
|----------|-------------|--------------|-----------------|-----------|-------------|------------|---------------------|---------------------|------------|
| 3        | 900         | 9            | In Transit      | 14        | 2           | 2023       | 0                   | 909                 | 14/2/2023  |
| 3        | 900         | 6            | Delivered       | 18        | 2           | 2023       | 0                   | 906                 | 18/2/2023  |
| 2        | 600         | 8            | In Transit      | 10        | 1           | 2023       | 0                   | 608                 | 10/1/2023  |
| 1        | 300         | 10           | Returned        | 30        | 3           | 2023       | 0                   | 310                 | 30/3/2023  |
| 5        | 1500        | 13           | Delivered       | 12        | 3           | 2023       | 0                   | 1513                | 12/3/2023  |
| 2        | 600         | 14           | Delivered       | 11        | 2           | 2023       | 0                   | 614                 | 11/2/2023  |
| 1        | 300         | 7            | Returned        | 18        | 12          | 2023       | 0                   | 307                 | 18/12/2023 |
| 1        | 300         | 7            | In Transit      | 14        | 1           | 2023       | 0                   | 307                 | 14/1/2023  |
| 2        | 600         | 17           | Delivered       | 16        | 4           | 2023       | 0                   | 617                 | 16/4/2023  |
| 1        | 300         | 15           | Delivered       | 5         | 4           | 2023       | 0                   | 315                 | 5/4/2023   |
| 1        | 300         | 7            | Delivered       | 26        | 10          | 2023       | 0                   | 307                 | 26/10/2023 |
| 3        | 900         | 18           | Delivered       | 20        | 7           | 2023       | 0                   | 918                 | 20/7/2023  |
| 5        | 1500        | 9            | In Transit      | 30        | 10          | 2023       | 0                   | 1509                | 30/10/2023 |
| 2        | 600         | 16           | Returned        | 2         | 9           | 2023       | 0                   | 616                 | 2/9/2023   |
| 3        | 900         | 17           | Returned        | 22        | 5           | 2023       | 0                   | 917                 | 22/5/2023  |
| 1        | 300         | 9            | Returned        | 25        | 7           | 2023       | 0                   | 309                 | 25/7/2023  |
| 1        | 300         | 17           | In Transit      | 23        | 11          | 2023       | 0                   | 317                 | 23/11/2023 |
| 2        | 600         | 18           | Returned        | 19        | 3           | 2023       | 0                   | 618                 | 19/3/2023  |
| 1        | 300         | 14           | Delivered       | 5         | 8           | 2023       | 0                   | 314                 | 5/8/2023   |
| 4        | 1200        | 15           | In Transit      | 28        | 1           | 2023       | 0                   | 1215                | 28/1/2023  |
| 1        | 300         | 17           | In Transit      | 3         | 4           | 2023       | 0                   | 317                 | 3/4/2023   |

The Order Date Column is created using Calculated Column in Dax Functions as Go to Table Tools Tab→ In that Select New Column (Calculations Column).

The Formula for Creating the Order Date,

OrderDate = CONCATENATE(CONCATENATE(CONCATENATE(ProductSales[Order Day],"/"),ProductSales[Order Month]),CONCATENATE("/","Year"))

The screenshot shows the Power BI Desktop interface. The 'Formulas' tab is active, displaying the DAX formula for the 'Sales Category' column: `Sales Category = If(ProductSales[Total Price] > 299,"Above Average","Below Average")`. The 'Data' pane on the right shows the 'ProductSales' table with columns: Shipping Fee, Shipping Status, Order Day, Order Month, Order Year, Order Time Duration, Total Product Price, OrderDate, and Sales Category. The 'Sales Category' column is highlighted in the 'Data' pane.

| Shipping Fee | Shipping Status | Order Day | Order Month | Order Year | Order Time Duration | Total Product Price | OrderDate  | Sales Category |
|--------------|-----------------|-----------|-------------|------------|---------------------|---------------------|------------|----------------|
| 9            | In Transit      | 14        | 2           | 2023       | 0                   | 909                 | 14/2/2023  | Above Average  |
| 6            | Delivered       | 18        | 2           | 2023       | 0                   | 906                 | 18/2/2023  | Above Average  |
| 8            | In Transit      | 10        | 1           | 2023       | 0                   | 608                 | 10/1/2023  | Above Average  |
| 10           | Returned        | 30        | 3           | 2023       | 0                   | 310                 | 30/3/2023  | Above Average  |
| 13           | Delivered       | 12        | 3           | 2023       | 0                   | 1513                | 12/3/2023  | Above Average  |
| 14           | Delivered       | 11        | 2           | 2023       | 0                   | 614                 | 11/2/2023  | Above Average  |
| 7            | Returned        | 18        | 12          | 2023       | 0                   | 307                 | 18/12/2023 | Above Average  |
| 7            | In Transit      | 14        | 1           | 2023       | 0                   | 307                 | 14/1/2023  | Above Average  |
| 17           | Delivered       | 16        | 4           | 2023       | 0                   | 617                 | 16/4/2023  | Above Average  |
| 15           | Delivered       | 5         | 4           | 2023       | 0                   | 315                 | 5/4/2023   | Above Average  |
| 7            | Delivered       | 26        | 10          | 2023       | 0                   | 307                 | 26/10/2023 | Above Average  |
| 18           | Delivered       | 20        | 7           | 2023       | 0                   | 918                 | 20/7/2023  | Above Average  |
| 9            | In Transit      | 30        | 10          | 2023       | 0                   | 1509                | 30/10/2023 | Above Average  |
| 16           | Returned        | 2         | 9           | 2023       | 0                   | 616                 | 2/9/2023   | Above Average  |
| 17           | Returned        | 22        | 5           | 2023       | 0                   | 917                 | 22/5/2023  | Above Average  |
| 9            | Returned        | 25        | 7           | 2023       | 0                   | 309                 | 25/7/2023  | Above Average  |
| 17           | In Transit      | 23        | 11          | 2023       | 0                   | 317                 | 23/11/2023 | Above Average  |
| 18           | Returned        | 19        | 3           | 2023       | 0                   | 618                 | 19/3/2023  | Above Average  |
| 14           | Delivered       | 5         | 8           | 2023       | 0                   | 314                 | 5/8/2023   | Above Average  |
| 15           | In Transit      | 28        | 1           | 2023       | 0                   | 1215                | 28/1/2023  | Above Average  |
| 17           | In Transit      | 3         | 4           | 2023       | 0                   | 317                 | 3/4/2023   | Above Average  |
| 18           | Returned        | 21        | 3           | 2023       | 0                   | 1518                | 21/3/2023  | Above Average  |

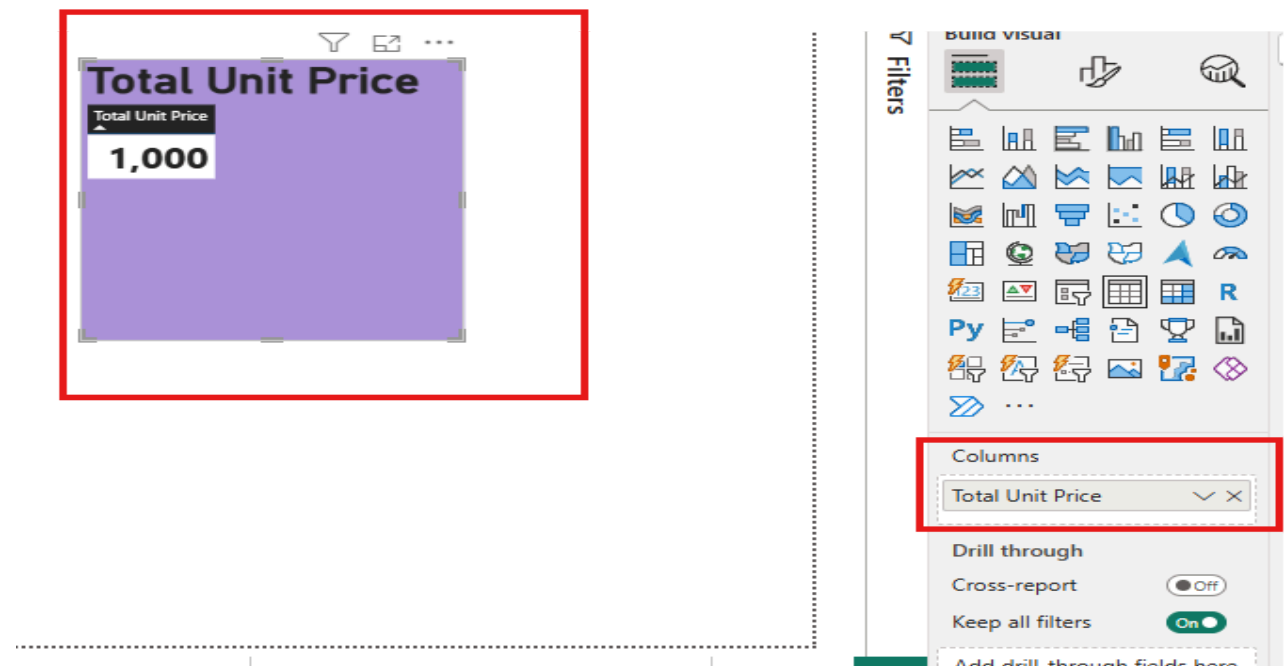
Sales Category Column have created using If Condition to show results as Data value >299, then the result is Above Average otherwise give Result as Below Average Which is done using the following formula as,

Sales Category = If(ProductSales[Total Price] > 299,"Above Average","Below Average")

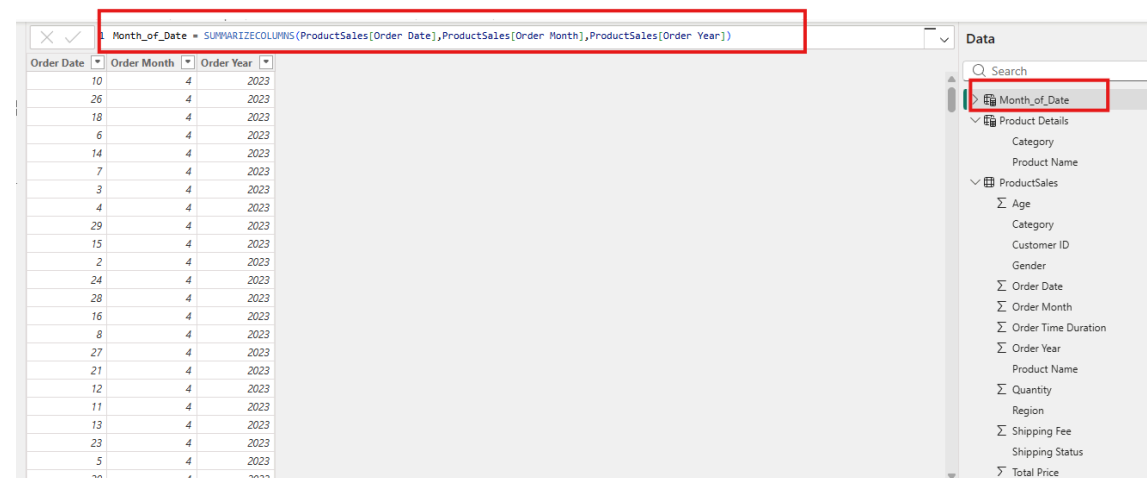
Calculated Measure:

Calculated Measure is a DAX Functions in Power BI, the calculation that can be applied to any visual. Here the New Measure is created for Total Unit Price

Total Unit Price = `COUNTA(ProductSales[Unit Price])`



Calculated Table:



Calculated Table is created to get a new table in Power Query DBI using DAX Functions.

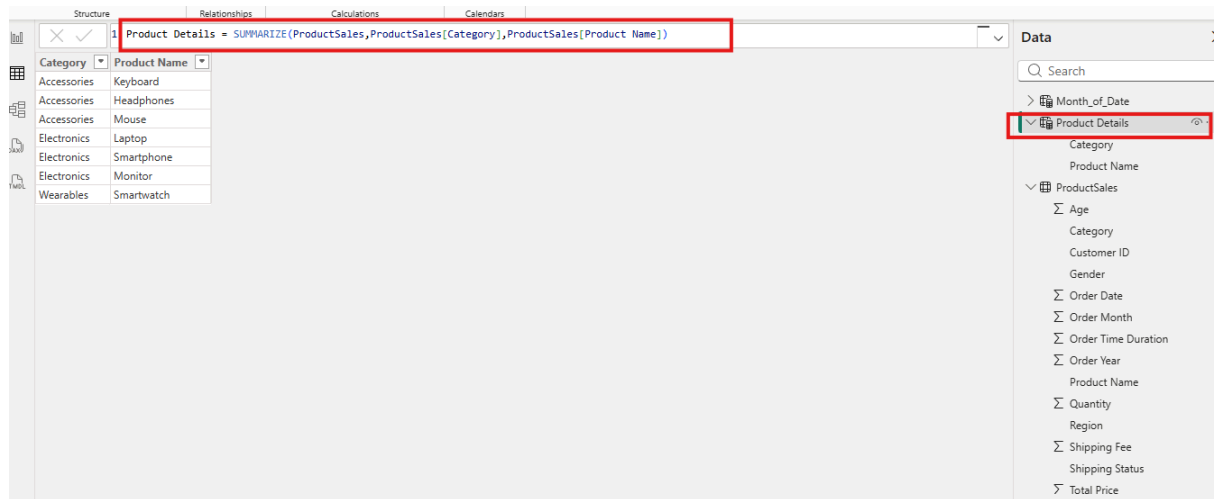
Go to Table tools→Calculation column select New Table as follows,

Month\_of\_Date = `SUMMARIZECOLUMNS(ProductSales[Order Date],ProductSales[Order Month],ProductSales[Order Year])`

For Establishing Relation Between two or more tables I have created this table to show Cardinality.

Product Details Table the Syntax for creating new table,

ProductDetails = `SUMMARIZE(ProductSales,ProductSales[Category],ProductSales[Product Name])`

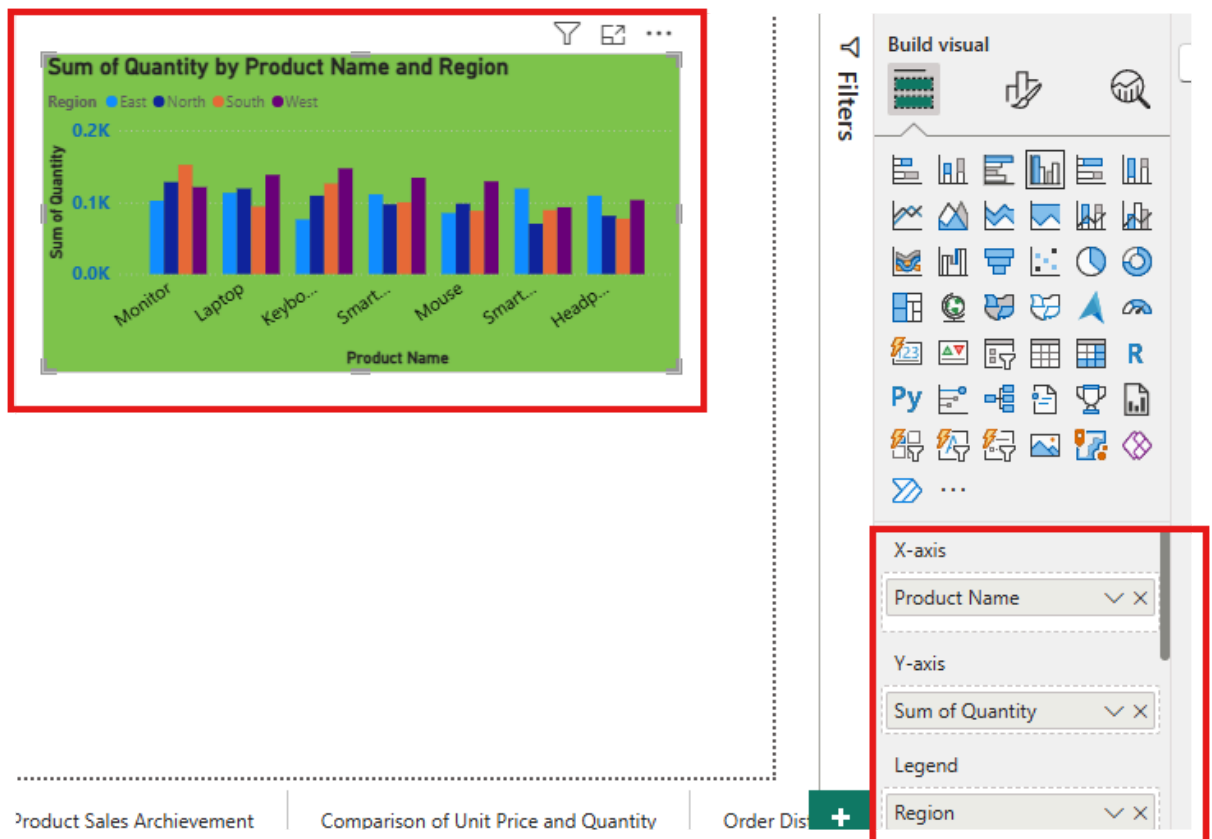


## 8. Analysis and Visualizations (Power BI)

### Dashboard Features:

- **Multiple Visualizations based on problem statements:** Bar charts, line charts, pie charts, cards, and tables to communicate insights.
- **Clustered Column Chart for Product Name by Region:**

A Clustered Column chart is used to compare categories by displaying grouped Columns for each category. The Clustered column chart is selected from the Visual pane→select Clustered Column chart, in that fill the x axis Categorical data and y axis Numeric values.

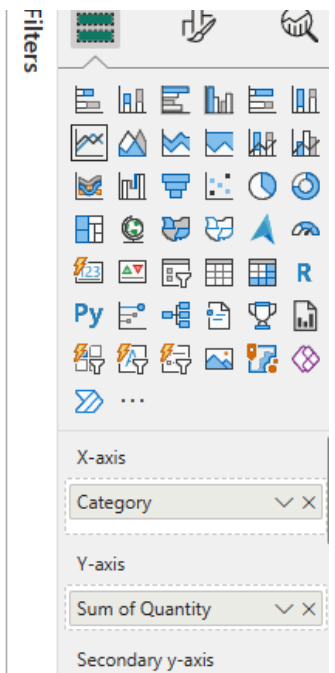
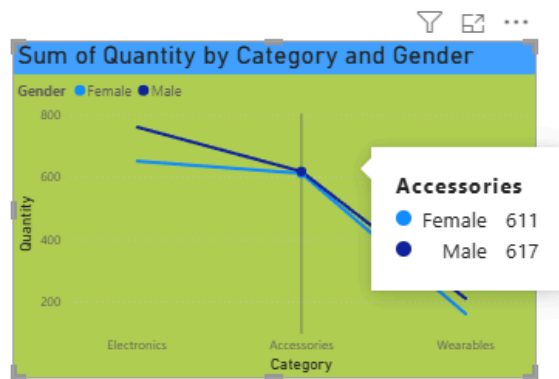


### Full Stacked Bar Chart:

It is used to show the Percentage contribution of each category where the total is 100%. here the category is contributed by Gender in which both male and female total 100% by Order Date.

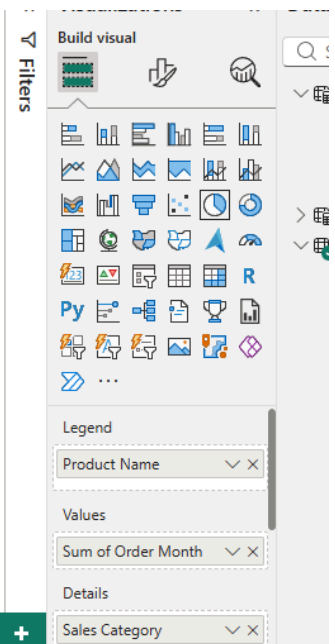


### Line Chart:



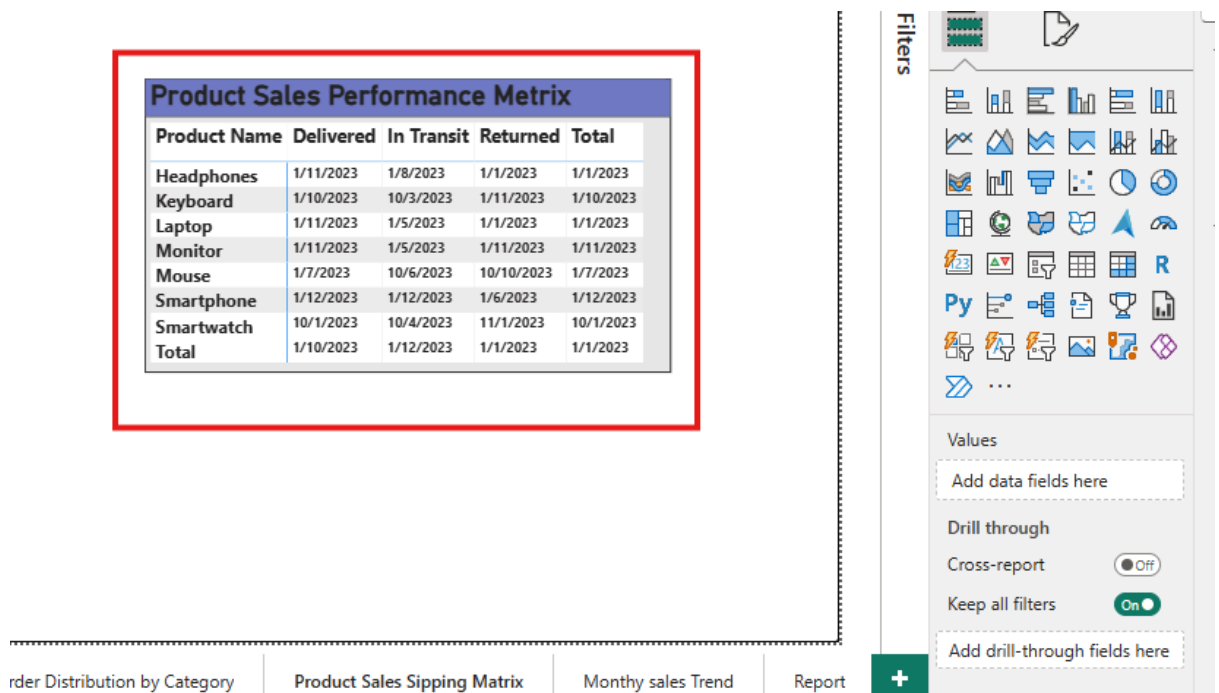
### Pie Chart:

Pie Chart shows visual Proportion of Category as a Whole. Here the visual shows the Monthly Trend for the Product sales according to sales category Below Average, Above Average by Product Names.



### Performance Metric:

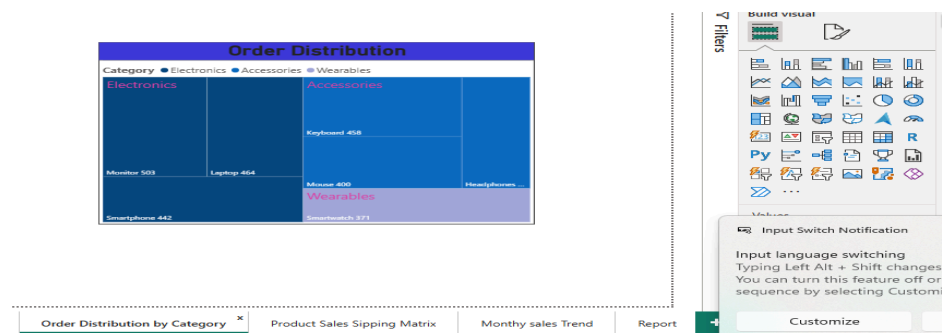
Order Delivery Details is Visualized here based on the Product, Shipping status



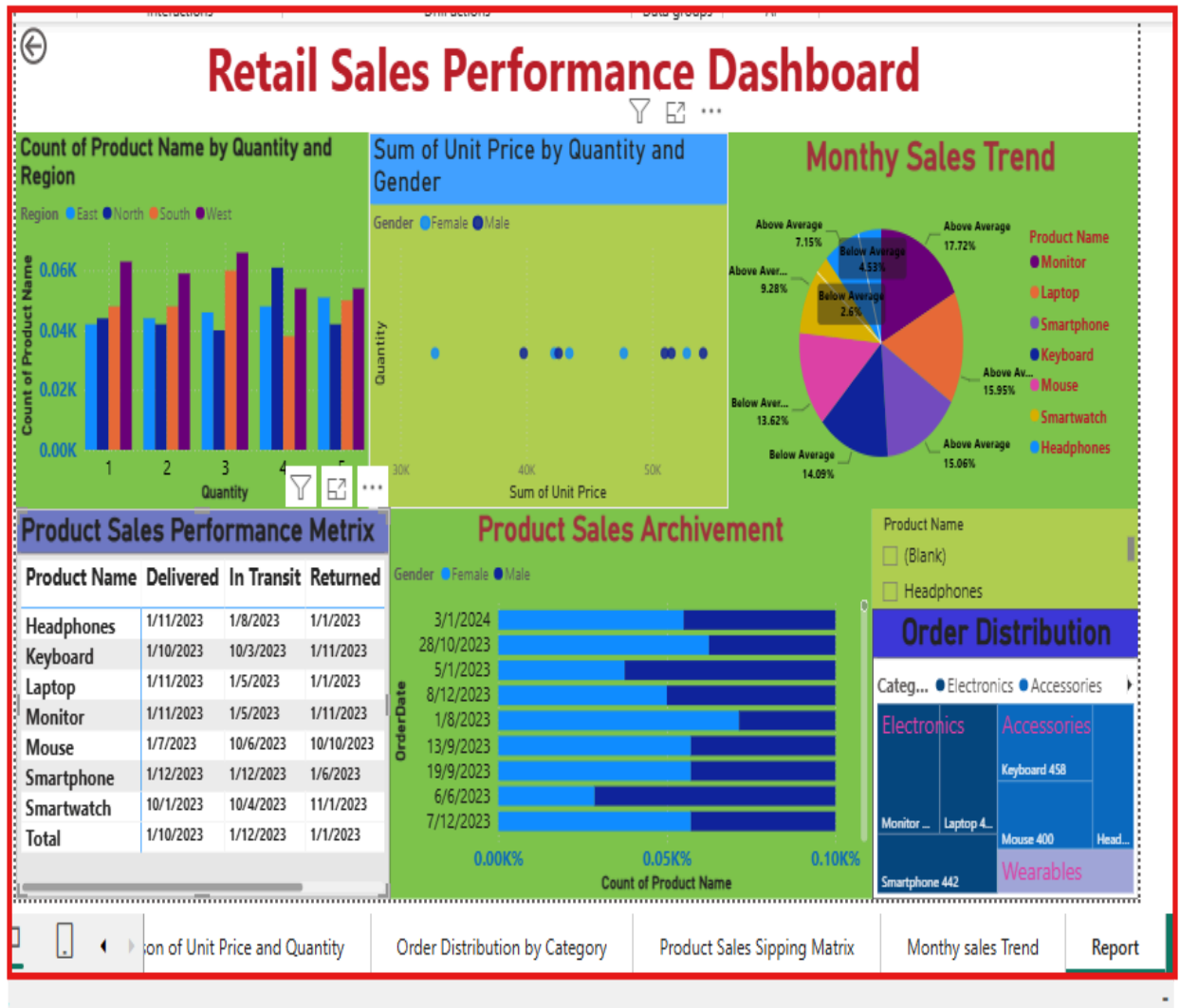
### Tree Map Chart:

Tree Map shows Data values or Performance High to Low ex.50,47,34,29,13likewise.

Here the tree map shows sales of Product Category order placed by the Customers.



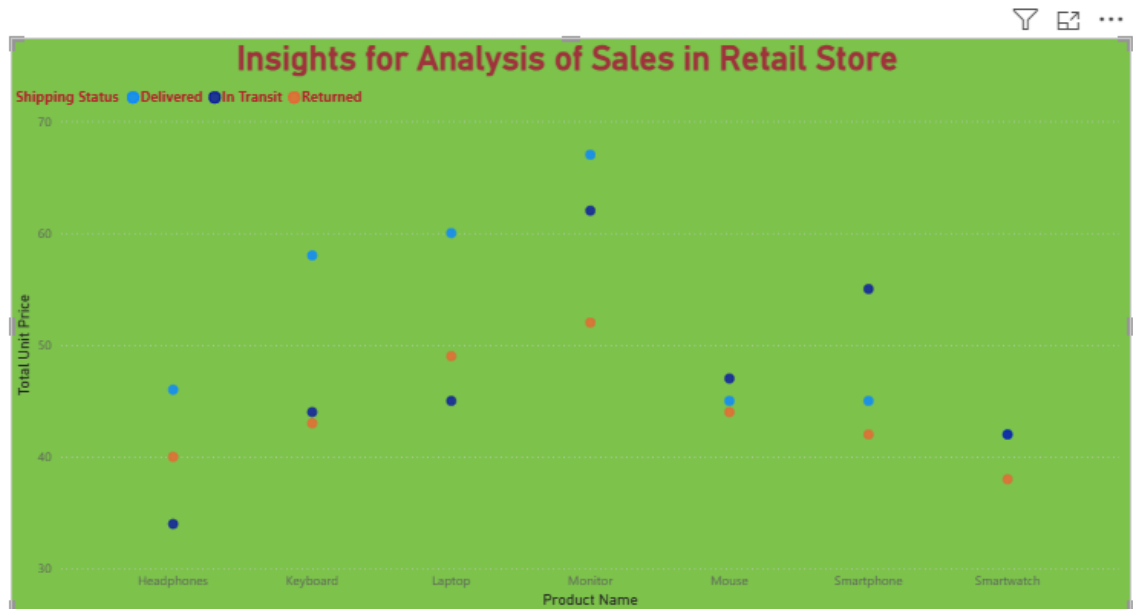
- **Report /Dashboard:**



## 9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data
- **Provide the analysis insights:**





- **Descriptive-Product** sale happened based on the customer order in multiple categories , here some inconsistency in the delivery part only.
- **Diagnosis-Following** the customer through any appliances is most important until the order gets delivered.
- **Predictive-In** future the retailer should focus on the demanded Product and invest in that more give high revenue.
- **Prescriptive-Increasing** Digital Platforms for Customers, to place order and easy to communicate with that Platforms.

## 10. Conclusions:

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting. The Reporting show various perspectives for sales Analysis and how Category of Product is Delivered to the customer and how to rectify some shipping status drawbacks to overcome form that. Hence the mini Project has been completed.